

DEEPAK VASUDEVAN

PL/SQL Developer · Data Engineer · Data Analyst · Power BI Developer

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■ PROFESSIONAL SUMMARY

PL/SQL Developer with 4.5+ years building Oracle-based data systems in a live hospital environment, now transitioning into Data Engineering and Analytics. Delivered two end-to-end production ETL pipelines processing 1,558,000+ patient records using Python, PySpark, Apache Airflow, dbt, and PostgreSQL. Proven at query optimization (40% performance gain), clinical KPI reporting, and automating reporting workflows (20+ hours/month saved). Seeking Data Engineer, Data Analyst, or PL/SQL Developer roles where I can scale data infrastructure and deliver measurable business value.

■ CORE TECHNICAL SKILLS

Languages	SQL · PL/SQL · Python 3.10 · Bash
Databases	Oracle 11g/12c/19c/21c · PostgreSQL 18
ETL & Pipelines	oracledb · Apache Airflow 2.9.1 · dbt Core 1.11 · SQLAlchemy
Big Data	PySpark 4.1.1 · Apache Parquet · Medallion Architecture (Bronze-Silver-Gold)
Data Modelling	Star Schema · Fact & Dimension Tables · Incremental Load · Watermark Pattern
Python Libraries	pandas · pyarrow · oracledb · python-dotenv · pyyaml · psycpg2
BI & Reporting	Power BI Desktop · KPI Dashboards · Analytical Data Cubes
Tools & DevOps	Git · GitHub · VS Code · Oracle SQL Developer · TOAD · pgAdmin
Domain	Healthcare — Hospital Information Systems · Clinical Data · Billing Analytics

■ WORK EXPERIENCE

Software Engineer — PL/SQL Developer · Mehta

Multispecialty Hospital India Pvt. Ltd., Chennai

Dec 2021 – Present (4.5 Years)

- Developed and optimized Oracle SQL/PL/SQL stored procedures, functions, triggers, and packages supporting clinical and finance reporting for a 200+ bed hospital
- Built 10+ SQL-based dashboards saving 20+ hours/month in manual reporting for finance and clinical stakeholders
- Improved query performance by 40% through index optimization and execution plan tuning; designed analytical cubes for budgeting, billing, and revenue analysis
- Built PL/SQL ETL workflows for reliable data extraction, transformation, and loading across hospital modules
- Automated DB access and audit processes reducing approval turnaround by 30%
- Performed source-to-target reconciliation and supported L2/L3 production issues within SLA timelines

■ PORTFOLIO PROJECTS — PRODUCTION ETL PIPELINES ON LIVE HOSPITAL DATA

Project 1 — Surgery Patient Journey ETL Pipeline

Stack: *Python · oracledb · pandas · pyarrow · dbt Core · PostgreSQL · Apache Airflow · Power BI*

- Extracted 125,871 real surgery records from live Oracle 19c using Python + oracledb thin mode (no Oracle Client installation required)
- Built Bronze-Silver-Gold medallion architecture with Parquet as intermediate storage — full raw data auditability and transform reproducibility
- Designed PostgreSQL star schema: 1 fact table + 6 dimension tables using dbt Core; calculated clinical KPIs including LOS (max 202 days), surgery duration (max 1,020 min), and age band grouping
- Implemented incremental daily load with watermark table for automatic failure recovery; deployed 5-task Airflow DAG with 2-retry logic and SLA monitoring at 5 AM
- Built data quality quarantine: 5 check types with rejection reason tagging; Power BI dashboard for surgeon caseload, LOS trends, and monthly discharge volumes

Project 2 — OP/IP Patient Registration ETL Pipeline

Stack: *Python · oracledb · PySpark 4.1.1 · pyarrow · dbt Core · PostgreSQL · Apache Airflow · Power BI*

- Unified 958,657 outpatient + inpatient records from two Oracle schemas (ip + admin) using UNION ALL extraction across 2018–2026 history
- Processed full historical load using PySpark 4.1.1 (8GB driver memory); loaded 1,558,430 rows into PostgreSQL via dbt with surrogate key generation in 25 seconds
- Implemented --mode FULL/DAILY CLI switch for single-codebase handling of initial load and daily incremental
- Deployed Airflow DAG running daily since April 2026; key insight: 849K OP vs 609K IP, Chetpet 666K vs Global 893K

■ EDUCATION

Degree	Institution	Year	Score
M.Sc. Information Technology	University of Madras (IDE)	2019–2021	8.0 CGPA
B.Sc. Computer Science	Dr. MGR Educational & Research Institute	2016–2019	7.4 CGPA

■ CERTIFICATIONS & CONTINUOUS LEARNING

- Power BI Workshop (AI-Powered Dashboards) — Office Master, 2025
- Introduction to Prompt Engineering — Simplilearn, 2024
- Actively upskilling in: Generative AI & Advanced Power BI, Cloud Fundamentals, Advanced PySpark & dbt